

Semiconductor Startup Landscape in India

*An Investor's Perspective
2025*



SPECIALE INVEST



Founded in 2014, Ankur Capital is an early-stage venture capital firm in India, investing in digital and deep science technology companies.

Ankur operates with a unique team, with a core of scientist-investors and entrepreneur-investors with global experience.

As recognized leaders in India's deep science tech, agri-tech and climate-tech domains, we understand the nuances of building in the Indian B2B ecosystem.

This report serves as a follow-up to the presentation *Semiconductor Startup Landscape in India – An Investor's Perspective* delivered by Sunil Cavale from **Speciale Invest** and Vishal Katariya from **Ankur Capital** at the 2nd Annual SemiX Summit, IIT Bombay, in November 2024.

The strong interest and engagement from both the organizers and participants at the 2024 Summit inspired this updated presentation and accompanying report in 2025.

For completeness, slides from the 2024 presentation have been included in the appendices.

This presentation was first presented at the [3rd Annual SemiX Summit at IIT Bombay](#) in September 2025.

All data and information are as of September 2025.

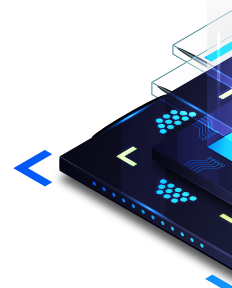
Key takeaways from the report

Following a 2024 high, funding in Indian semiconductor startups has remained steady in 2025 – investments have been fewer in number but more meaningful in size.

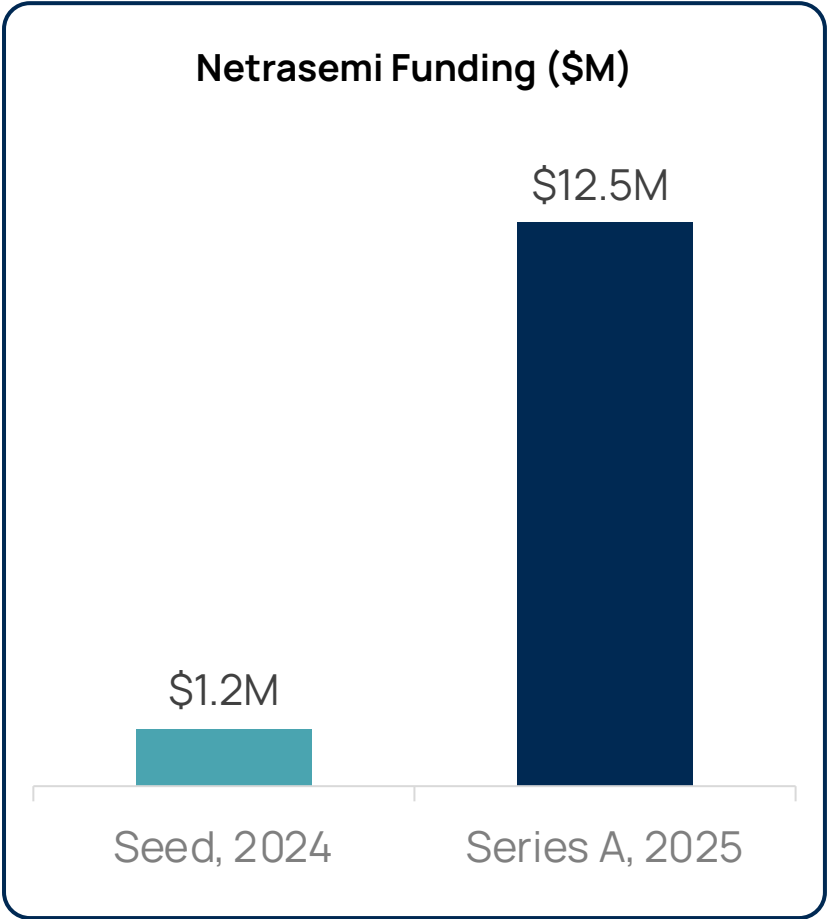
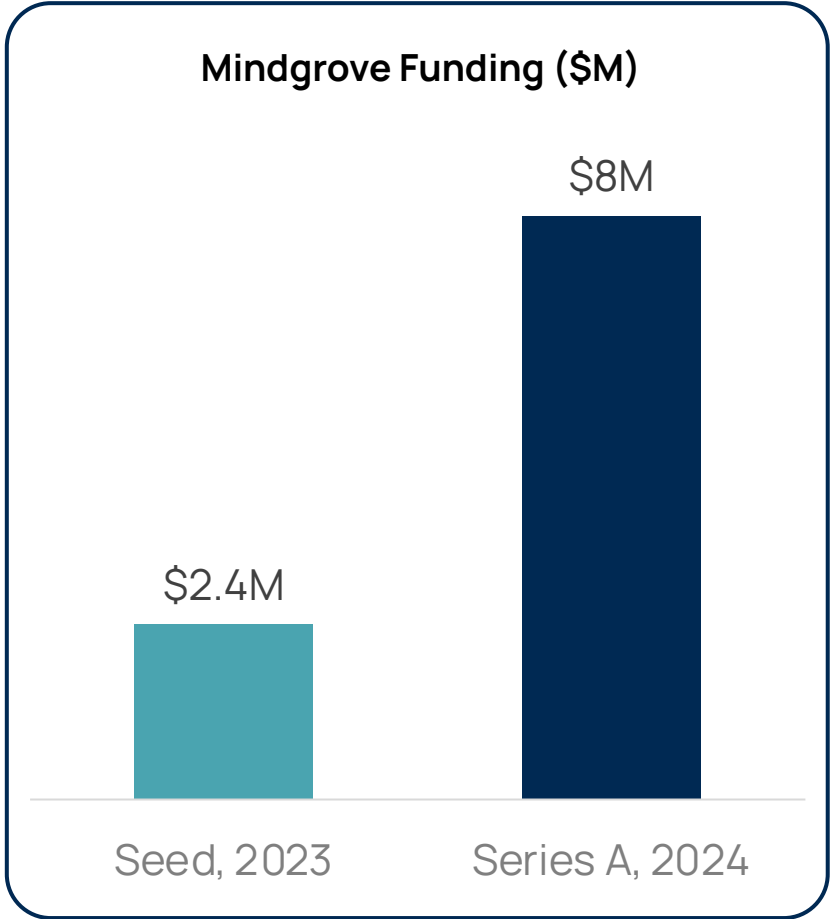
Three companies (Mindgrove Technologies, Morphing Machines, & Netrasemi) raised their Series A rounds within 18 months of their seed rounds.

Notable observations

- [Large seed rounds at the idea stage driven by founder-led conviction](#) (Agrani Labs, Maieutic Semi)
- [Continued trend of second-time entrepreneurs](#) (Steradian founding team starting Maieutic Semi with a large seed round)
- [Emergence of Indian customers for Indian semiconductor chips](#) (marked by product partnerships)
- [The rise of startups building equipment and tools for semiconductor manufacturing](#)
- [Fresh focus from large listed companies to invest into the semicon design ecosystem](#), including product and design services companies



Significant Series A rounds within 18 months of seed rounds, outlier in the Indian landscape



Continued trend of second-time entrepreneurs

2024

TESSOLVE

IVP
SEMI

2025

STERADIANSEMI

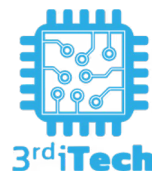
maieutic

In 2024 and 2025, we saw Raja Manickam and Ashish Lachhwani start new semicon startups after successful exits from their previous companies.

Significant product partnerships in 2025



Mindgrove-designed chips to be used for automotive, consumer electronics & IoT applications. The chips will have Bosch contributions on R&D + software development.



A nine-member Indian consortium led by Kaynes Semicon will supply over 500,000 fully indigenous dash and surveillance cameras by March 2026.

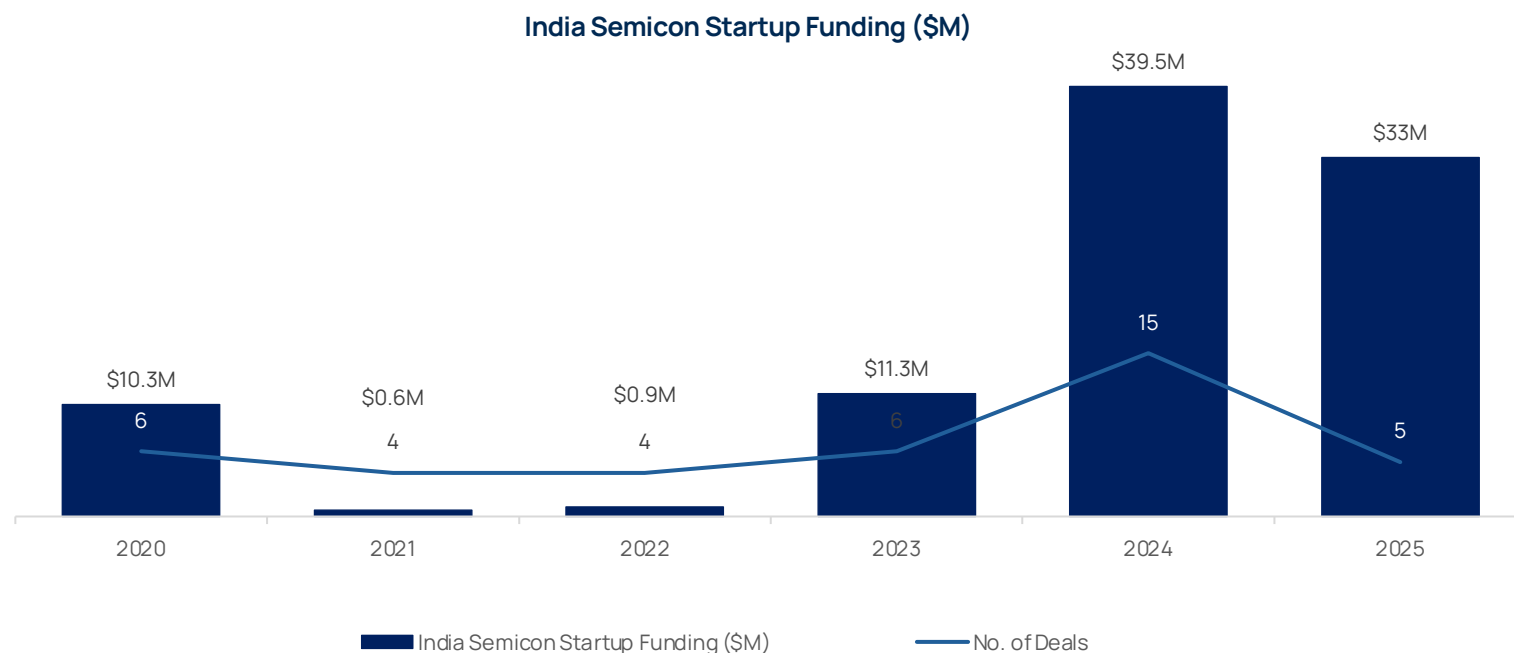


Boat to co-develop semiconductor chips with HrdWyr with Tata Electronics set to be the assembly partner.



FermionIC Design partnered with Tata Electronics for India's first 4-Channel X-Band Beamformer IC. The partnership includes wafer-level testing, packaging, & post-silicon validation.

2025 funding remains robust, dominated by larger investment rounds



Source: Tracxn. Funding information includes design, product, IP and manufacturing tech startups in the Indian semiconductor ecosystem

Highlights

We've seen sizeable A rounds this year as well as large conception-stage seed rounds.

Design services co. Tessolve raised \$150M from TPG for inorganic growth and M&A.

Kinara, US-India chip design startup acquired by NXP in \$307M deal.

2025 funding has been marked by significant Seed and Series A rounds

Series A fundraises



\$12M

Zoho, Unicorn
India Ventures



\$8M

Rocketship VC,
Mela Ventures

Seed fundraises



\$6M

Investors
unannounced

maieutic

\$6M

Endiya Ventures,
Exfinity Ventures,
UTEC

Acquisitions headlined by Kinara.ai



Kinara AI, a US-India semiconductor startup, was acquired by NXP for a landmark \$307M, significantly larger than other 2019-2024 acquisitions.

maieutic



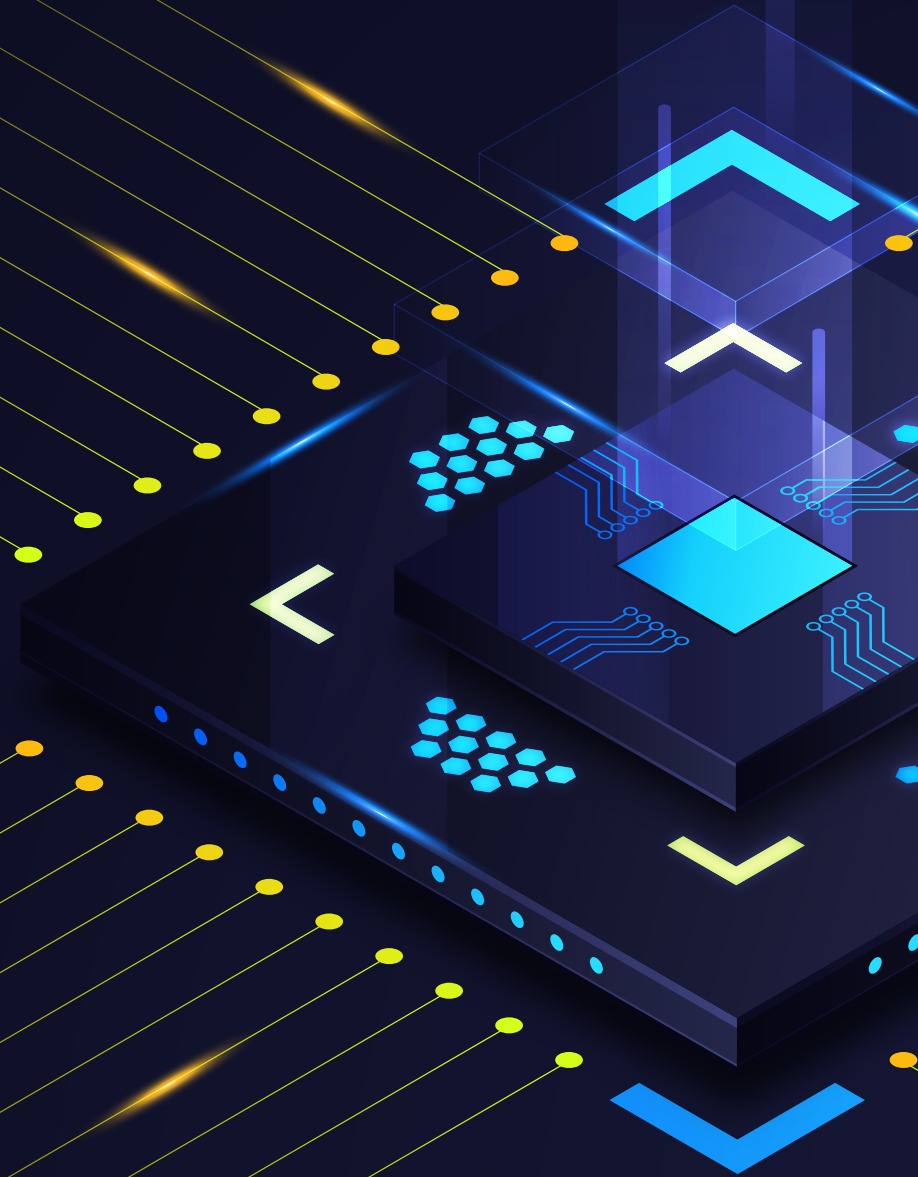
Endiya

Exfinity (after the Kinara acquisition) & Endiya (first investors in Steradian) invested in Maieutic Semi, co-founded by the core Steradian team.

Year	Acquired company	Acquirer	Acquisition Value
2024	Renesas' RF components unit	CG Power	\$36M
2024	G.G.Trionics	CG Power	\$38M
2024	SiliConch	L&T Semiconductor	\$16M
2024	Cientra	Accenture	-
2024	Excelmax	Accenture	-
2024	Insemi	Infosys	\$33M
2022	P2F SEMI	Tessolve	-
2022	SteradianSemi	Renesas Electronics	-
2022	Saankhya Labs	Tejas Networks	\$58M
2020	Cerium Systems	Tech Mahindra	\$32M
2019	Sankalp Semi	HCL Technologies	\$25M

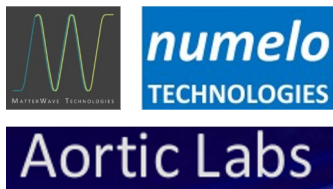


Trends we saw in 2025



Activity continues in fresh academic spin-offs

Maturing spin-offs



Semiconductor startups from India's tier 1 institutes have raised seed and Series A rounds in 2024.

Fresh 2025 spin-offs



This has catalysed new ventures being started or spun off in 2025.



Investment activity is beginning to move beyond fabless chip design models

Metrology and
quality inspection



Lithography



Process equipment
& innovations



Low volume
manufacturing



MICROBEAR APPLIED SCIENCES PRIVATE LIMITED

Throughout 2023 and most of 2024, both policy drive and investor interest focused on [fabless startups](#).

2025 marked a change on that front, with a wave of manufacturing and tooling startups.

From simpler chips to AI GPUs



2025 saw the rise of startups with vision stretching beyond simple chips to graphics processing units (GPUs) and AI accelerators.

These startups

- ✓ raised \$4M+ conception-stage seed rounds on the back of strong founding teams;
- ✓ had founding teams come from both established semicon companies & new-age inferencing startups; and
- ✓ are targeting both global and Indian markets from the get-go.

Large Market Opportunities → Strong Startup Activity

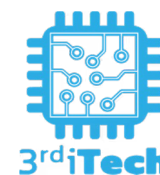
Data Centres

The data centre landscape is evolving - from new [HPC](#) and AI accelerator chips to [interconnect](#), security and cooling technology. There are many fabless design, [photonics](#) and materials science startups in this space.



Cameras

Supported by on-device AI, intelligent cameras are becoming more capable. Bolstered by a robust domestic market, a cohort of startups are building intelligent, secure and connected cameras for defence, surveillance and automotive applications.



Emerging trends in Capital and Partnerships

01



Looking beyond venture capital -

Startups such as FermionIC, Suchi Semicon & Netrasemi have accessed equity funding beyond venture capital funds.

2025 saw new domestic venture capital funds take positions in semiconductor startups.

02



Large listed companies in India

(Cyient, Hero Group, Tata Electronics) getting into the semiconductor space are looking to go full stack and partner with IP/design services companies.

These companies are also emerging as **new strategic investors** into semiconductor startups.

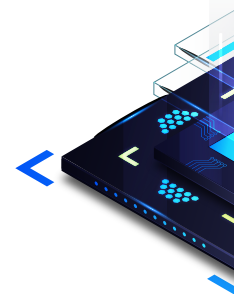
03



Maturing product-focused

partnerships between startups and Indian packaging players, for example, Lightspeed / Kaynes, and 3rdiTech / Sensesemi / Kaynes.

What investors look for in early-stage semicon startups



Founders with domain expertise

- ✓ Strong founding teams with domain expertise – usually with both hardware and software expertise.
- ✓ CEOs/founders who have past background in engaging with customers and can handle business conversations.
- ✓ Having a strong set of advisors to guide on product development, market access etc.

Customer-focused development

- ✓ Start conversing with potential customers early to understand what the requirements are. This is not to sell your product, but more to understand the pulse of the customer. Product development and business exploration should happen in parallel.
- ✓ If your customers are in the US, prioritize spending time there at least 3-4 times a year.

What investors look for in early-stage semicon startups

Focus on the right use-case



While an IP/chip can solve for multiple use cases, it would be good to have a narrow focus and understand where the beachhead market opportunity is and why that is the one to pursue.

Grants and Partnerships



Partnerships that give you access to EDA tools and licenses.

Leverage Design Linked Incentive (DLI), Chips to Startup (C2S) and other grants that are available.

Fundraise plan tied to milestones



First fundraising should take you to a multi-project-wafer (MPW) tapeout when applicable.

If you have successfully taped out a chip already, that adds to the investor confidence and potential valuation premiums.

Have a pragmatic view on what will get you your next fundraising.

India continues to remain an exciting place to build

Continuing **relocalization, reshoring and geopolitical** pressures on the semiconductor ecosystem

There is flux across **tariffs, talent & immigration**

- As early-stage investors, tariffs don't affect our views or investment behaviour too much
- From our perspective, these cause some short-term pain, and some long-term opportunities
- We look forward to the Indian talent ecosystem being bolstered by overseas talent

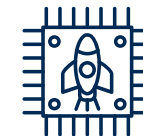
Sharpening focus of Indian semiconductor + startup manufacturing ecosystem

- Continuing progress on foundries and packaging/testing plants (OSATs) being built
- Ecosystem feedback being incorporated into Design Linked Incentive (DLI) 2.0
- Ecosystem for compound semi, displays, etc. is growing to support startups and MSMEs
- “Full-stack” consortiums being built with clear product + customer focus



What are we excited about for 2026?

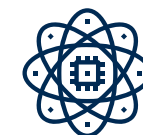
DLI 1.0 startups coming out with new products, raising next institutional rounds and getting chips in the hands of customers.



DLI 2.0: We are looking forward to the refreshed DLI scheme, with larger and more diverse capital outlay, restructured support framework to support beyond design and test chips; and for it to be a catalyst for fresh activity.



More **“full-stack” product consortiums** to cater to specific market areas, bringing startups, industry and government together.



Changing landscape of design services as design shops consolidate to increase utilization and garner growth investor interest, while also ascending the skill & product ladder themselves.





If you are an entrepreneur, investor, or expert in this space,
email me. We would love to learn about your work, or have
your input in our next report.

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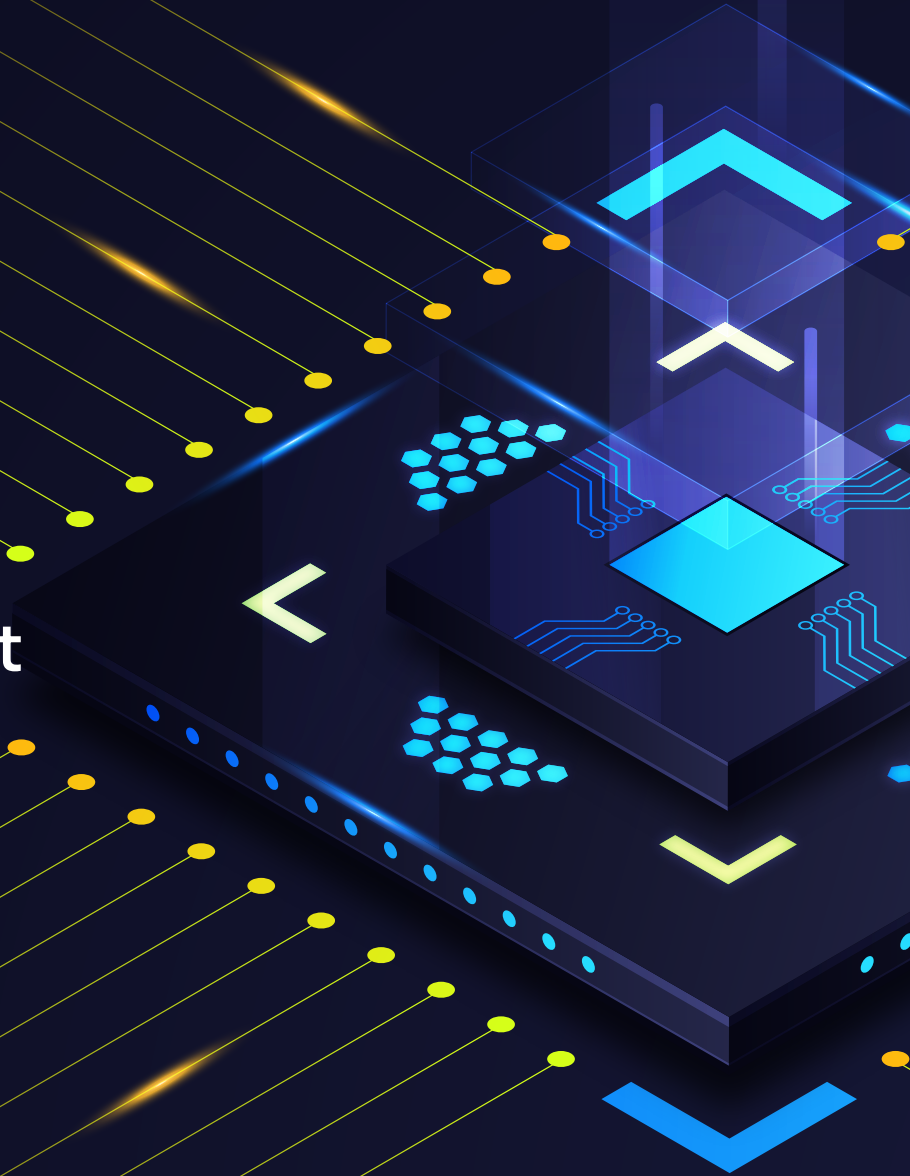
Authors

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Thanks for reading.



Appendix: 2024 Semiconductor industry report



Global Tailwinds for Semiconductors

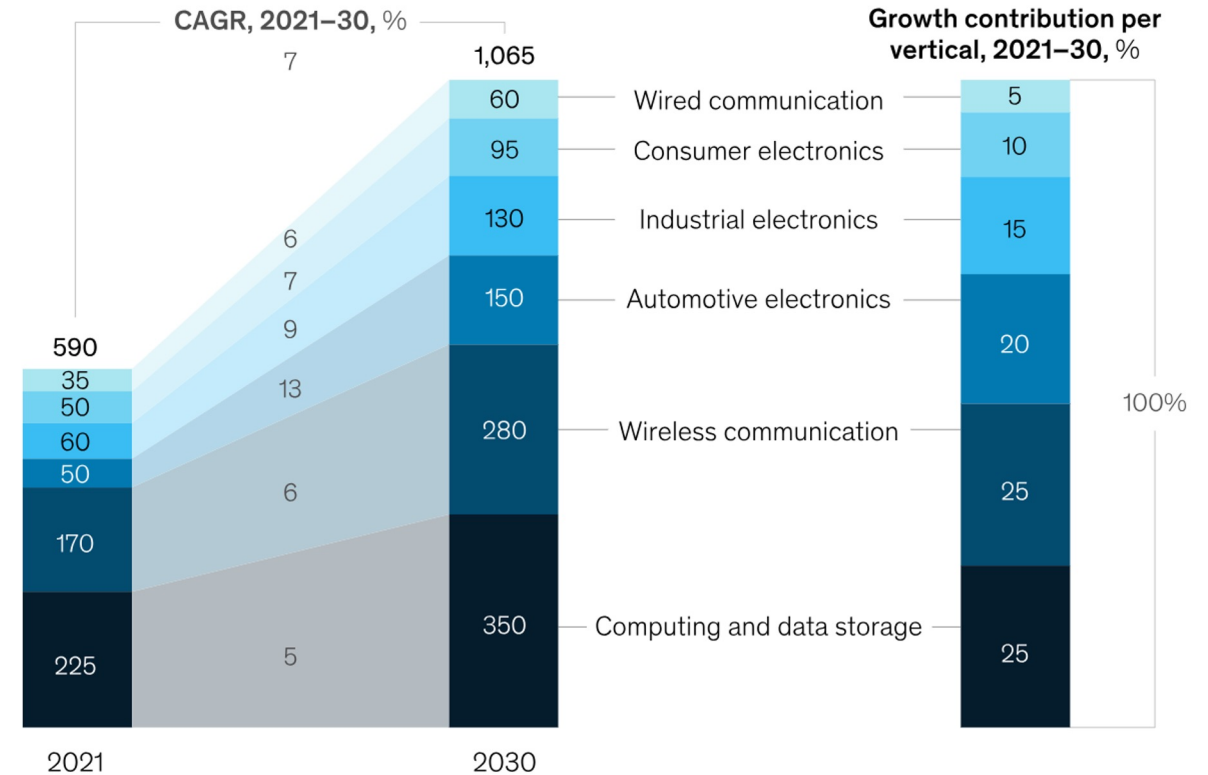
Ever-growing demand: Semiconductor industry is already at \$600B and is growing at 15% per year.

New age applications coming up across IoT, Wireless, data centres, automotive, AI etc and pushing for new innovations and opportunities for startups.

Geopolitics / supply chain interdependencies: re-shoring and de-centralizing of critical semiconductor resources supported by Govt incentives such as the CHIPS Act, ISM.

Competencies cannot be developed overnight – either at a micro or a macro level; tech moats built in.

Global semiconductor market value by vertical, indicative, \$ billion



Source: McKinsey & Company, April 2022

Renewed Focus on Semiconductor Startups in India

Government funding for R&D – DLI, C2S, along with tangential funding like TTDF, iDEX.

Large emerging markets in India - Design in India, build from India for India – large markets in IoT, defence, vision, EVs which did not largely exist before.

Design in India, **build in India, for the world** – Indian talent and startups building technologies for new-age applications data centres, Edge, AI etc..

Talent availability – 20% of the global design talent is based out of India, **which has matured to be more holistic** – looked at tape outs, product, customer engagement, sales etc..



The infographic is titled "Design Linked Incentive (DLI) Scheme" and is part of the "Make in India Semiconductors for the World" initiative. It is issued by the Ministry of Electronics & Information Technology, Government of India, and features the myGov logo. The scheme details are as follows:

Fiscal Support	
Product Design Linked Incentive: Reimbursement Up to 50% of the eligible expenditure subject to a ceiling of ₹15 crore	Deployment Linked Incentive Incentive of 6%-4% of net sales turnover over 5 years subject to a ceiling of ₹30 crore
Tenure 5 years starting from 01.01.2022	Beneficiaries 100 Domestic semiconductor design companies

The infographic also includes icons for a stack of coins with an upward arrow, a calendar with a checkmark, and a blue semiconductor chip.

Strong Capital Outlay Powering India as a Global Hub for Electronics

\$10B

Support for Semiconductor and Display Ecosystem

- Semiconductor Fabs and Display Fabs
- Compound Semiconductor and ATMP
- Design Linked Incentive (DLI)
- Modernization of Semiconductor Laboratory (SCL)

\$8.5B

Support for Electronics Manufacturing

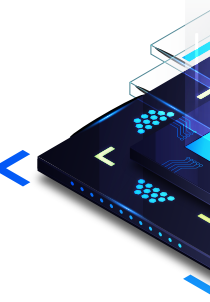
- Production Linked Incentives for Mobile Phones, Components, IT Hardware
- Capex Linked Incentives for components, sub-assemblies
- Development of Electronics Manufacturing Clusters

\$13B

Support for Allied Sectors

Production Linked Incentives for:

- Advanced Cell Chemistries
- Automobiles and Auto Components
- Telecom and Networking
- Solar PV Modules
- Medical Devices



Diving into the \$10 Billion Outlay for Semiconductor Development

The 2024 Union Budget saw a 71% hike in funding for chip and electronics manufacturing.

Semicon Fabs

Fiscal Support: 50% of project cost on pari-passu basis

Technologies covered: All node sizes for Silicon CMOS based fabs

Display Fabs

Fiscal Support: 50% of project cost on pari-passu basis

Technologies covered: New-gen TFT LCD & AMOLED displays

Compound Semicon & Packaging

Fiscal Support: 50% of CAPEX pari-passu basis

Technologies covered: OSAT/ATMP, Compound Semiconductor, Silicon Photonics (including MEMS)

Semicon Design

Product Design Linked Incentive: 50% of eligible expenditure

Design Infrastructure Support

Founder Personas



Founders are coming out of large global design organizations to build products on their own – coming back with strong experience in design, product, customers, sales etc Ex: OptoML, aiSIC, ABCRL, LineSemi, Makr.



TEXAS
INSTRUMENTS

intel®

APPLIED
MATERIALS®

Qualcomm



Professors and Researchers building startups based on research done in academia.



AGNIT

numelo
TECHNOLOGIES



Founders moving back to India from abroad looking to build in India.

SHCON
SOLUTIONS

Levi



Second-time founders in semiconductor space – moving from services to products.

IVP
SEMI

Ecosystems Supporting Semicon Startups

Semicon specific accelerators/CoEs allowing for focussed support to semicon startups.



Private companies supporting acceleration of semicon startups.



Indian diaspora in the US offering insights into the US market and getting access to customers.



Corporate venture arms of semicon majors are spending more time in India.



DLI/C2S are expected to follow a similar path as BIRAC's BIG grants to provide early support to semicon startups.

